Augmenting Machine Learning with Genetic AI

A Hybridisation Approach: Layers, Attention, and Open Directions

Summary. We investigate whether the inductive biases of Genetic AI, specifically the fixed-point fitness computation that models population-level gene competition, can enrich standard machine learning components. Rather than replacing gradient-based learning, we treat genetic fitness as a structural prior that can be hybridised into neural networks at different levels of granularity. This white paper reports two completed experiments and two open research directions. *Experiment I* hybridises genetic fitness into feed-forward and multi-head layers across seven tabular classification benchmarks. *Experiment II* modulates value projections in BERT attention with genetic fitness for dense passage retrieval on MS-MARCO. The central finding is that hybridisation succeeds in *parallel* multi-head topologies, where genetic competition acts as an ensemble prior, but degrades in deep sequential stacks. Genetic priors also accelerate early training dynamics before saturating at convergence, pointing to adapter and routing architectures as the most promising future direction.

1. Introduction

Genetic AI derives its computational structure from biological evolution: populations of candidate solutions (organisms) and their features (genes) compete via a fitness function. When rendered as a differentiable fixed-point computation, mapping a population matrix $\mathbf{P} \in [0,1]^{n_{\text{org}} \times n_{\text{gene}}}$ to per-gene and per-organism fitness scores, this mechanism becomes a *structural prior* that can be inserted into neural network architectures at specific points. Unlike purely evolutionary approaches that compete with gradient descent, the hybridisation approach explored here *augments* learned transformations: the genetic computation is fixed-formula and parameter-free, while the surrounding projections are trained normally with standard optimisers.

This document reports the findings of two hybridisation experiments and proposes two further directions, ordered by the architectural component targeted:

1. **Genetic Layers** (§3): replacing feed-forward and multi-head linear transformations with GeneticLayer modules across seven tabular classification benchmarks.
2. **Genetic Attention** (§4): modulating value projections in BERT attention with genetic fitness for dense passage retrieval.
3. **Genetic Adapter** (§5): proposed multi-task classification on a frozen encoder.
4. **Genetic MoE** (§5): proposed fitness-based expert routing in MoE architectures.

2. Background: Genetic Fitness as a Structural Prior

Population Matrix and Fitness Computation

The genetic prior is expressed through a *population matrix* $\mathbf{P} \in [0,1]^{n_{\text{org}} \times n_{\text{gene}}}$, obtained from a neural network's intermediate activations via a sigmoid projection. Each row represents an organism (a candidate solution or feature group) and each column a gene (a latent dimension). Gene fitness is then computed as:

$$\tilde{\phi}_g = \frac{1}{n_{\text{org}}} \sum_{i=1}^{n_{\text{org}}} \mathbf{P}_{ig}, \tag{1}$$

$$\gamma_g^* = \left(\sum_{s=1}^{n_{\text{gene}}} \frac{\tilde{\phi}_g + \frac{1}{2}}{\tilde{\phi}_s + \frac{1}{2}} \right)^{-1}. \tag{2}$$

Organism fitness follows as:

$$\boldsymbol{\phi}_{\text{org}} = \mathbf{P}\,\boldsymbol{\gamma}^*. \tag{3}$$

The $+\frac{1}{2}$ offset prevents division by zero and moderates competition when genes are similarly expressed. Eq. (2) is a *fixed-formula* computation: no additional parameters are learned beyond the projection matrices that produce $\mathbf{P}$. This is the core property that makes Genetic AI suitable as a structural prior, it introduces biologically-motivated inductive bias (relative gene competition) without adding learnable capacity, relying instead on the surrounding gradient-trained layers to shape the population matrix from which fitness emerges.

3. Experiment I: Genetic Layers

Background

A standard feedforward layer maps input $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$ through a learned linear transformation followed by a non-linearity σ:

$$\mathbf{h} = \sigma(W\mathbf{x} + \mathbf{b}), \qquad W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}},\ \mathbf{b} \in \mathbb{R}^{d_{\text{out}}}.$$

A multi-head network runs K independent heads in parallel and concatenates their outputs via an output projection W_O:

$$\text{MH}(\mathbf{x}) = \big[\mathbf{h}_1(\mathbf{x}) \parallel \cdots \parallel \mathbf{h}_K(\mathbf{x})\big] W_O.$$

Both topologies are fully parametric: all weight matrices are learned by gradient descent. This experiment replaces the core non-linear transformation with the genetic fitness computation (Eqs. (1)–(3)). Concretely, a genetic layer wraps the fixed-formula fitness computation between two learned linear projections: an input projection (LayerNorm → Linear → SiLU) maps the input to a hidden dimension, a genetic

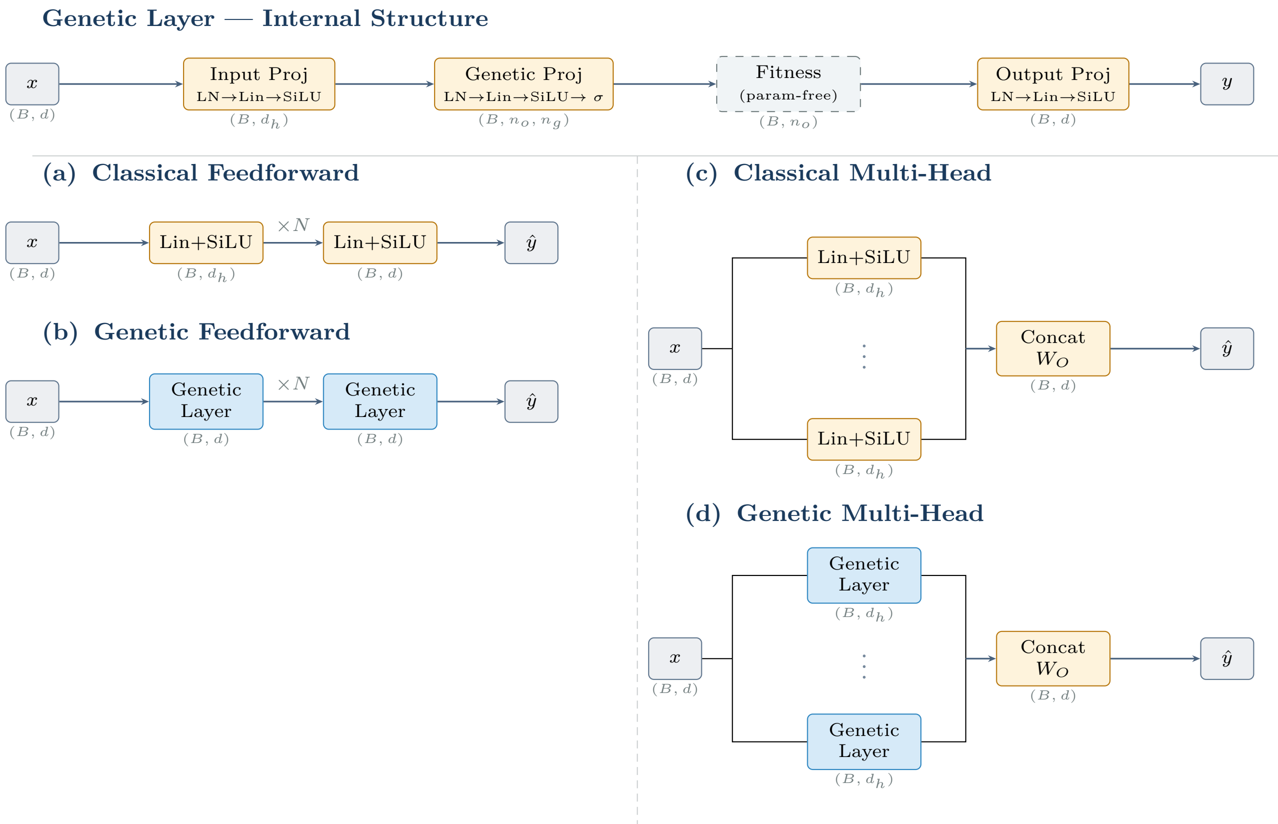

Figure 1: *Top*: internal structure of a Genetic Layer (Section 3). Amber boxes are learned projections (LayerNorm → Linear → SiLU); dashed grey box is the parameter-free fitness computation. *Bottom*: the four topology variants evaluated in Experiment I. Amber boxes are all learned blocks (linear layers and output projections); blue boxes are Genetic Layer modules (each expands $(B,d) \to (B,n_o,n_g) \to (B,n_o)$ then projects back to (B,d)). Labels show output tensor shape (B = batch, d = feature dim, d_h = head/hidden dim, n_o = organisms, n_g = genes).

projection (LayerNorm → Linear → SiLU → sigmoid) maps that to the population matrix $\mathbf{P} \in [0,1]^{n_{\mathrm{org}} \times n_{\mathrm{gene}}}$, the fitness computation then produces organism fitness scores, and a final output projection maps those to the target dimension. The learned weights are in the projections; the fitness computation itself has no parameters. Figure 1 shows the internal structure of a Genetic Layer and the four topology variants.

Setup

The first hybridisation experiment asks whether the genetic fitness prior can substitute for fully learned linear transformations in a feed-forward classification network. We replace the linear layers in two standard network topologies with hybrid *genetic layers*: each layer projects its input to a population matrix via a sigmoid, applies the fitness computation of Eqs. (1)–(3), and returns the organism fitness vector. The surrounding projection weights are trained with Adam; the fitness computation itself is parameter-free. Seven tabular classification datasets are used, spanning 4 to 4,096 input features (Iris, Wine, Breast Cancer, Digits, Synthetic, CovType, Olivetti Faces). Two topologies are evaluated: *feedforward* stacks (depths 2, 4, 6) and *parallel heads* (1, 2, 3 heads concatenated). All configurations: 128 epochs, Adam optimiser, batch size 16,384, early stopping. The evaluation metric is balanced accuracy.

Table 1: Average performance across all seven datasets.

Topology	**Type**	**Bal. Acc.**	**Params**
Feedforward	Classical	88.7%	48,851
Feedforward	Genetic	49.1%	25,667
Heads	Classical	79.5%	41,330
Heads	Genetic	**80.8%**	44,235

Results

The topological split is stark and consistent. Genetic feedforward networks underperform their classical counterparts by 39.6 percentage points on average and, crucially, *degrade with depth*: on the Digits dataset a genetic feedforward achieves 53.0% balanced accuracy at depth 2, but only 29.4% at depth 4 and 18.8% at depth 6, the inverse of the monotone improvement seen for classical networks (94.8%, 97.2%, 97.5%). The same pattern holds on every high-dimensional dataset. Because Eq. (2) is a fixed-formula transformation, stacking multiple such layers does not build abstraction; it compounds information loss at each stage.

Multi-head topologies tell a different story. Genetic heads outperform classical heads on average (80.8% vs. 79.5%) and exhibit a *stronger benefit from increasing head count*: classical head networks show only moderate improvement as heads increase from 1 to 3, whereas genetic head networks improve substantially. Table 2 shows the three datasets with the largest genetic advantage.

These three datasets share medium feature dimensional-

Table 2: Datasets where genetic heads most exceed classical heads (best genetic head count vs. best classical head count).

Dataset	Genetic	Classical	Δ
Synthetic (50 feat.)	59.6%	46.8%	+12.8pp
CovType (54 feat.)	75.0%	66.6%	+8.4pp
Digits (64 feat.)	97.6%	94.8%	+2.8pp

ity (50–64) and multi-class output structure (7, 7, and 10 classes). We speculate that parallel genetic heads learn complementary fitness landscapes: because each head maintains its own Input and Genetic Projection weights, the same input x is mapped to a distinct population matrix $\mathbf{P}$ per head, and the fixed-formula fitness (Eq. (2)) is then applied to each of these independently learned representations. This induces an ensemble effect across diverse population views, whereas a single deep stack repeatedly applies the same transformation and collapses toward an uninformative fixed-point.

Parameter efficiency is competitive in the multi-head setting: genetic heads use only 7% more parameters than classical on average, compared to 48% *fewer* for genetic feedforward (which achieves far less accuracy). Training overhead is notable: 17% slower for genetic heads and 27% slower for genetic feedforward; a meaningful cost that warrants consideration in compute-constrained settings.

Limitations

These are preliminary results. Each configuration was evaluated on a single fixed train/test split (80/20, `random_state=8080`) with no cross-validation, and each model was trained once with no repeated runs. For small datasets such as Iris (n=150) and Wine (n=178), a single split can produce estimates with high variance, and reported differences may not survive a change of seed. The depth-degradation finding (Section 3) is consistent enough across all seven datasets to be robust, but the per-dataset multi-head advantage figures in Table 2 should be treated as indicative rather than conclusive.

Key Takeaway

Genetic AI hybridisation is **topology-sensitive**: it fails in sequential depth stacks and succeeds in parallel multi-head configurations. Hybrid multi-head networks match or outperform fully-learned counterparts on several medium-dimensional multi-class benchmarks at a comparable parameter budget.

4. Experiment II: Genetic Attention

Background

Multi-Head Attention (MHA) [6] maps the input $X \in \mathbb{R}^{B\times T\times d}$ (batch B, sequence length T, hidden size d) to queries, keys, and values $Q, K, V \in \mathbb{R}^{B\times T\times d_k}$ via learned projections $W^Q, W^K, W^V \in \mathbb{R}^{d\times d_k}$, with $d_k = d/h$ for h heads:

$$Q = XW^Q, \quad K = XW^K, \quad V = XW^V.$$

The attention output is the value-weighted sum over all positions, with weights derived from scaled query–key similarity:

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V.$$

All projection matrices are fully learned. The parallel-head success in Experiment I motivates asking whether a genetic prior on the value pathway can improve retrieval, keeping the QK similarity computation intact.

Design

The design was developed iteratively: if genetic competition across heads improves classification (Experiment I), can analogous competition improve retrieval when applied inside the attention mechanism? Initial attempts targeted a GPT-style decoder-only (autoregressive) architecture with causal masking, exploring two ways to inject genetic fitness:

- **Value-only**: Q and K are eliminated. The value matrix $V \in \mathbb{R}^{B\times h\times T\times d_k}$, scaled to $[0, 1]$, is used directly as the population matrix $\mathbf{P}$ (organisms = tokens, genes = d_k head features). Eqs. (1)–(2) yield gene fitness $\boldsymbol{\gamma}^* \in \mathbb{R}^{d_k}$; the score $\phi_{ij} = V_j \cdot \boldsymbol{\gamma}^*_i$ replaces the usual $Q_i \cdot K_j$ similarity.
- **Score-matrix modulation**: a $T \times T$ score matrix S is constructed (either from dedicated projections $O'G'^\top/\sqrt{d_k}$ replacing QK entirely, or as $\sigma(QK^\top/\sqrt{d_k})$ on top of the standard path). S is used as the population matrix (organisms = query tokens, genes = key tokens); Eqs. (1)–(3) yield per-token organism fitness $\boldsymbol{\phi}_{\text{org}} \in \mathbb{R}^T$, which re-weights the rows of S before softmax and multiplication by V.

Neither approach converged reliably in the decoder setting. Two compounding failure modes were identified:

Statistical instability. The gene-mean computation (Eq. (1)) estimates each gene's typical expression level by averaging normalised value activations across all sequence positions. In a decoder with causal masking, position t can only see positions $1, \ldots, t$, so early tokens average over just one or two values, which is an extremely noisy estimate. Because each position sees a different prefix length, $\tilde{\Phi}_g$ is also asymmetric across positions, making it an unstable learning signal during early training. The genetic prior is intended to be a stable structural signal; in the causal setting it becomes a moving, position-dependent quantity.

Computational cost. A correct causal implementation cannot compute a single shared gene-fitness vector per forward pass. To preserve causality, the fitness computation must be run separately for each prefix length $t = 1, \ldots, T$, producing T independent $\boldsymbol{\gamma}^*$ vectors before the attention output at any position can be formed. This makes the per-step cost scale as $O(T)$ gene-fitness evaluations rather than one, an overhead that compounds the instability problem and makes training impractically slow at even moderate sequence lengths.

Final Design

Both failure modes trace back to a fundamental structural mismatch: the gene-mean computation requires a stable average over *all* visible positions, but causal masking makes that set grow with each token, rendering the mean noisy and position-dependent. The genetic prior is therefore a

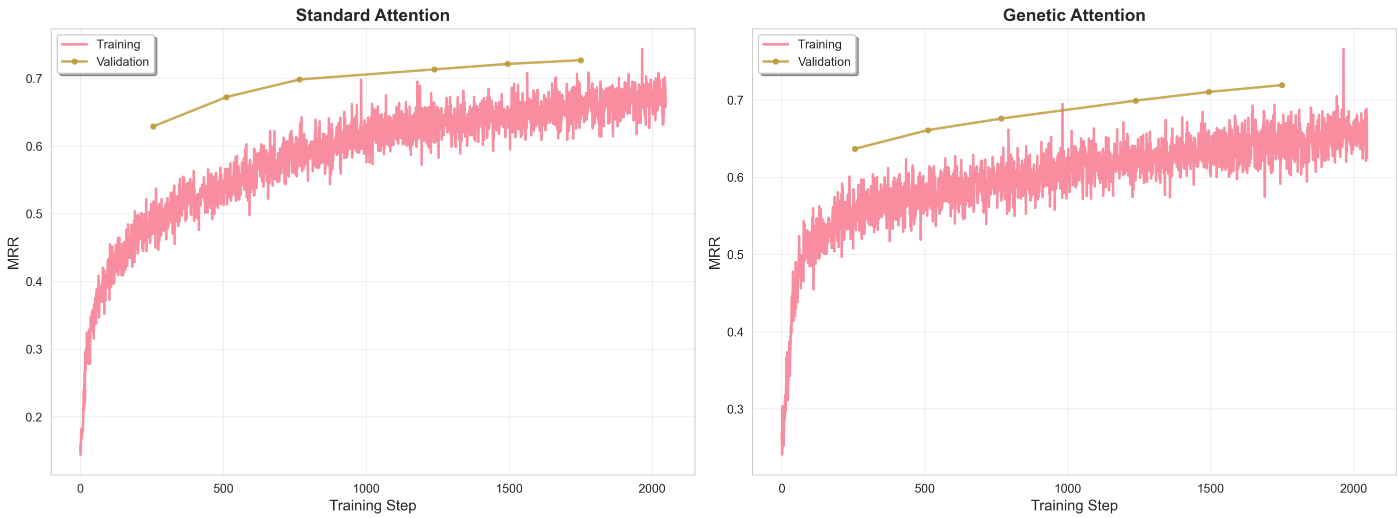

Figure 2: Mean Reciprocal Rank over training steps. Genetic attention leads early before being overtaken by the standard baseline, which continues improving to convergence.

natural fit for *non-causal* (bidirectional) attention, where every position sees the full sequence and a single, stable $\tilde{\Phi}_g$ can be computed per forward pass. We therefore applied the design to a BERT-style encoder, keeping the standard Q–K path intact and applying the genetic prior only to V. We call the resulting mechanism Multi-Head Genetic Attention (MGA):

$$V_{\text{norm}} = \sigma(V), \tag{4}$$

$$\tilde{\Phi}_g = \frac{1}{T}\sum_{t=1}^{T} V_{\text{norm}}[:,:,t,g], \tag{5}$$

$$\gamma_g^* = \left(\sum_{s=1}^{d} \frac{\tilde{\Phi}_g + \frac{1}{2}}{\tilde{\Phi}_s + \frac{1}{2}}\right)^{-1}, \tag{6}$$

$$V_{\text{gen}} = V \odot \boldsymbol{\gamma}^*, \tag{7}$$

$$\text{MGA}(Q,K,V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V_{\text{gen}}. \tag{8}$$

This modulates *which features* the attention output emphasises, without altering *which positions* it weights. The fitness $\boldsymbol{\gamma}^*$ is computed fresh for each forward pass from the value activations of the current batch, making it input-dependent and parameter-free beyond the existing value projection weights.

Setup

A six-layer BERT [3] (hidden size 384, 12 attention heads, $d_{\text{head}} = 32$, intermediate size 1,536) is trained as a dual encoder [4] for passage retrieval on MS-MARCO [1] (503K training pairs, 102K test pairs). Query and passage encoders share the same backbone. Sentence representations are obtained by mean pooling over valid tokens and L2-normalising. Training uses InfoNCE loss with in-batch negatives (batch size 512), AdamW with linear warmup, fixed seed 8080, for 2048 steps.

Results

At 2,048 steps, standard attention outperforms genetic attention across all retrieval metrics, and achieves a substantially lower best validation loss (2.42 vs. 2.98). However, the training curves (Figure 2) tell a more nuanced story: during the first several hundred steps, genetic attention *leads on retrieval metrics*, after which the standard baseline overtakes it and continues to improve while genetic attention plateaus. This early advantage is consistent with the hypothesis that $\boldsymbol{\gamma}^*$ provides a structured initialisation prior that front-loads useful feature selection, but once the value projections are sufficiently trained, the unconstrained attention has greater capacity to specialise. At convergence the genetic constraint acts as a ceiling rather than a scaffold.

Table 3: MS-MARCO dual-encoder results at 2,048 training steps.

Config	MRR	NDCG@5	Hit@5	Time (s)
Standard	0.492	0.500	0.573	7,822
Genetic	0.486	0.492	0.568	11,947
Δ	−1.2%	−1.6%	−0.9%	+52.8%

The 52.8% training overhead arises because $\boldsymbol{\gamma}^*$ is not a fixed parameter: it is derived from the value activations of the current batch on every forward pass. This means the full fitness computation (Eqs. (1)–(2)) runs as an extra sequential step after the value projection, on top of the standard attention computation, for every training step.

Limitations

This experiment was a single exploratory run per configuration (one fixed seed, 8080; one fixed train/val/test split of MS-MARCO). No repeated runs, hyperparameter sweeps, or significance testing were performed. The 2,048-step budget is intentionally short, sufficient to observe the early-training dynamic but not to characterise convergence behaviour fully. A rigorous comparison would require multiple seeds, runs to full convergence (tens of thousands of steps), and paired statistical tests. The reported metric differences (≤1.6 percentage points) are within a range that could plausibly be explained by seed variance alone.

Hybridising attention with a genetic value prior provides an **early-training advantage** in retrieval metrics but is overtaken at convergence. Fully-learned attention achieves lower validation loss and better test metrics with 53% less training time.

5. Future Directions

Experiment III: Genetic Adapter

The multi-head success in Experiment I motivates using genetic population heads as *classification adapters* on top of a frozen pre-trained encoder, rather than as a replacement for its internal components. We propose attaching a genetic multi-head module to a frozen BERT backbone and evaluating on the Bias-in-Bios dataset [2] (biography text, profession and gender labels). The multi-task structure provides a natural test of whether independent genetic heads partition fitness landscapes by task. Counterfactual evaluation via pronoun swapping would measure whether the organism-competition dynamic incidentally suppresses spurious gender correlations, a potentially valuable fairness property beyond raw accuracy.

Experiment IV: Genetic Expert Routing in MoE

MoE architectures [5] route each input token to a subset of expert feed-forward blocks via a learned router. We propose replacing this router with a genetic fitness computation: each expert is treated as an organism, and each token's feature dimensions as genes. The top-k experts are then selected by organism fitness (3). This would yield a parameter-free, biologically-motivated routing mechanism. The hypothesis is that genetic competition naturally implements diversity-preserving selection, analogous to the multi-head gains in Experiment I, without the load-balancing instabilities associated with learned routers. Validating this requires fine-tuning a model with at least $\sim$1B parameters and is deferred to higher-compute settings.

6. Conclusion

The two completed experiments establish a clear boundary condition for Genetic AI hybridisation with machine learning. First, hybridisation is *topology-sensitive*: inserting the genetic prior into sequential depth stacks compounds information loss at each fixed-formula stage, causing performance to degrade monotonically with depth. By contrast, parallel multi-head architectures benefit from the genetic prior because each head converges to a distinct fitness equilibrium, producing a complementary ensemble. Second, genetic priors *accelerate early training but saturate*: in the attention hybridisation experiment, genetic value modulation leads on retrieval metrics during the first several hundred steps before the unconstrained baseline catches up at convergence.

Together, these findings show potential advantages for Genetic AI hybridisation: the prior is most valuable when *diversity is incentivised* (e.g. parallel independent heads) and *training is short or cold-start-critical* (e.g. warm-up settings). They also show potential limitations: sequential depth and long fine-tuning, where the fixed-formula constraint becomes a ceiling rather than a scaffold. The proposed Genetic Adapter and Genetic MoE experiments are designed to test these findings at larger scale and with more complex task structure, with the goal of establishing whether genetic hybridisation can deliver reproducible gains beyond the early-training window.

All code and extended results with accompanying analysis documents are available at danube.ai's open-source repository for Genetic AI: `https://github.com/danube-ai/pikaia`.

Acronyms

BERT Bidirectional Encoder Representations from Transformers

MGA Multi-Head Genetic Attention

MHA Multi-Head Attention

MoE Mixture of Experts

NDCG Normalised Discounted Cumulative Gain

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